A Retrieval-Augmented Generation pipeline combines document processing, vector search, and language model generation.

The first stage is ingestion. A user uploads a document such as a PDF or TXT file. The application extracts readable text from the document and splits the text into smaller chunks.

The second stage is embedding generation. Each text chunk is sent to an embedding model. The embedding model converts the text into a numerical vector. In this project, the embedding model produces 384-dimensional vectors.

The third stage is vector storage. The generated vectors are stored in Qdrant, a vector database. Each vector is stored together with payload metadata such as the chunk text, chunk index, and character count.

The fourth stage is semantic retrieval. When a user asks a question, the question is also converted into an embedding. Qdrant compares the question vector with the stored chunk vectors and returns the most relevant chunks.

Top-K retrieval means returning the best matching chunks. For example, Top-K 5 means Qdrant returns the five most relevant chunks.

The final stage is answer generation. The retrieved chunks are passed to a large language model as context. The LLM uses this context to generate a grounded answer based on the uploaded document.